

Calculating PUF-PAS Sampling Rates and Effective Volumes in RStudio

A Step-by-Step Workflow

PUF-PAS Effective Volume Model

This R project calculates congener-specific sampling rates and effective sampling volumes for all 209 PCB congeners.

Safa Kalteh and Andres Martinez

IIHR-Hydrosience & Engineering
Department of Civil & Environmental Engineering
University of Iowa

Original Article

Web-Based Data Analytics Framework for Polyurethane Foam Passive Air Sampling Rate and Effective Volume

Nicholas Herkert^{1,2}, Bekir Zahit Demiray³, Ibrahim Demir^{4,5}, Andres Martinez², and Keri C. Hornbuckle^{2,*,†}

Abstract

Polyurethane foam equipped passive air samplers are a valuable tool for monitoring atmospheric semivolatile organic pollutants, but are often complicated due to the need for an accurate sampling rate (R_s). Here, we describe the development and implementation of a web-based R_s interface that allows for a user-friendly way of determining and visualizing accurate modeled R_s values. The web interface provides access to the previously published model without the need for proprietary software (MATLAB) or coding experience. The precalculated R_s avoids unnecessary computation time, providing the user with fast results and visualizations of the sampler behavior during deployment. Supported by the National Aeronautics and Space Administration's Modern Era Retrospective-Analysis for Research and Applications meteorological data at both 2 and 10 m heights above ground level, users can determine R_s for any location on the globe. Users can select prepopulated compounds (e.g., polychlorinated biphenyls) or manually enter compounds, enter sample information individually or upload them batch-wise, run the model, and export data to a CSV to further integrate with the data analysis workflow. Additionally, an implementation of the same model is available in R, along with scripts to process meteorological data.

Keywords

data analytics, effective sampling rate, effective sampling volume, modeling, PUF-PAS, web interface

¹ Nicholas School of the Environment, Duke University, Durham, North Carolina, USA.

² Department of Civil and Environmental Engineering and IIHR-Hydrosience and Engineering, University of Iowa, Iowa City, Iowa, USA.

³ Interdisciplinary Graduate Program in Informatics and IIHR-Hydrosience and Engineering, University of Iowa, Iowa City, Iowa, USA.

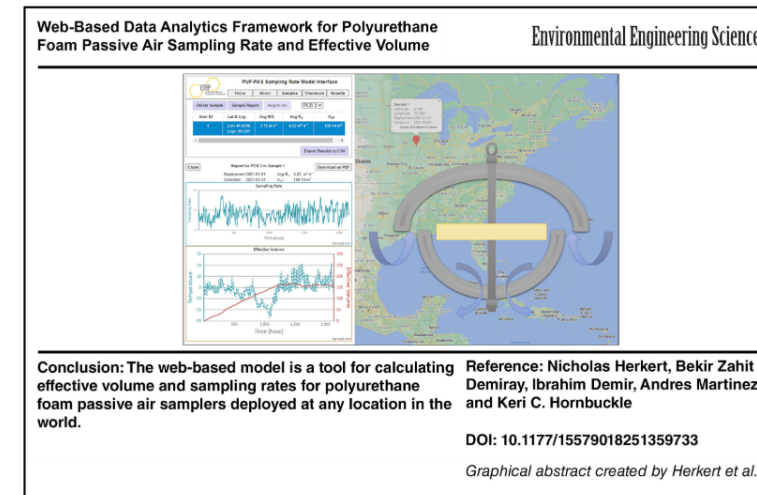
⁴ River-Coastal Science and Engineering, Tulane University, New Orleans, USA.

⁵ ByWater Institute, Tulane University, New Orleans, USA.

Corresponding author(s):

* Corresponding author: Department of Civil and Environmental Engineering and IIHR-Hydrosience and Engineering, University of Iowa, Seamans Center for the Engineering Arts and Sciences, Iowa City, IA 52242, USA. keri-hornbuckle@uiowa.edu

† AEESP member.

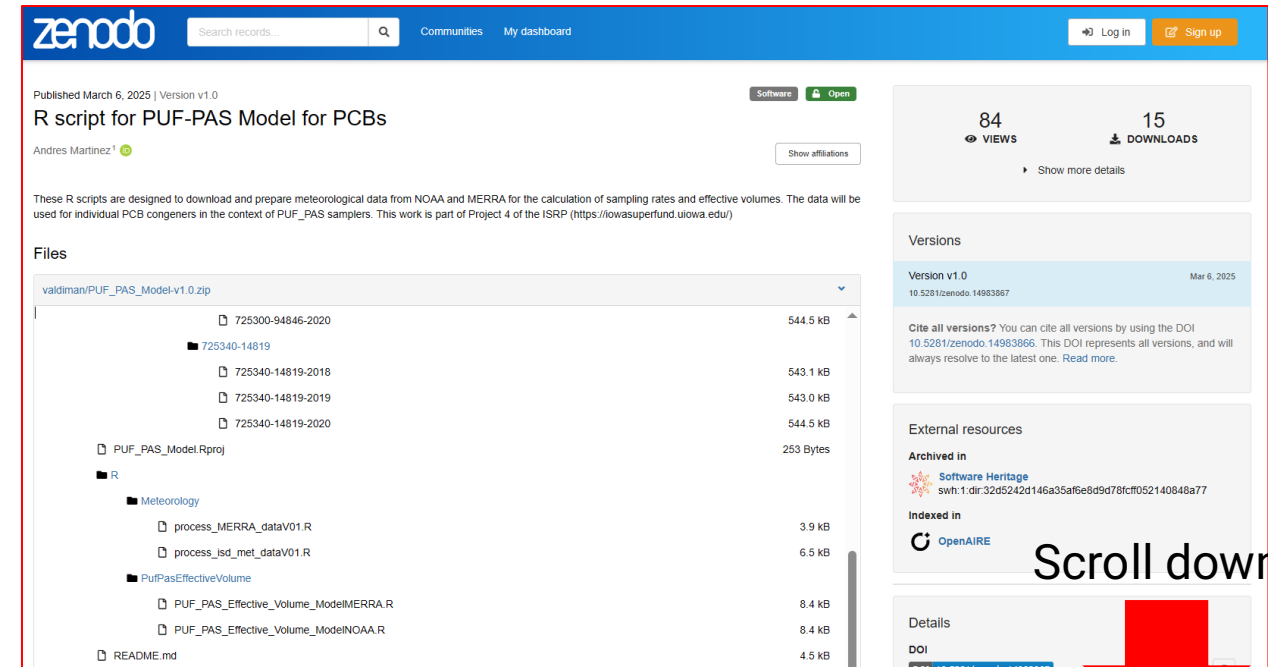


R-project Contents

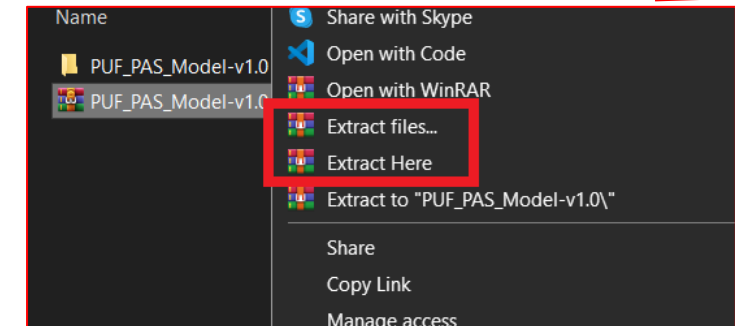
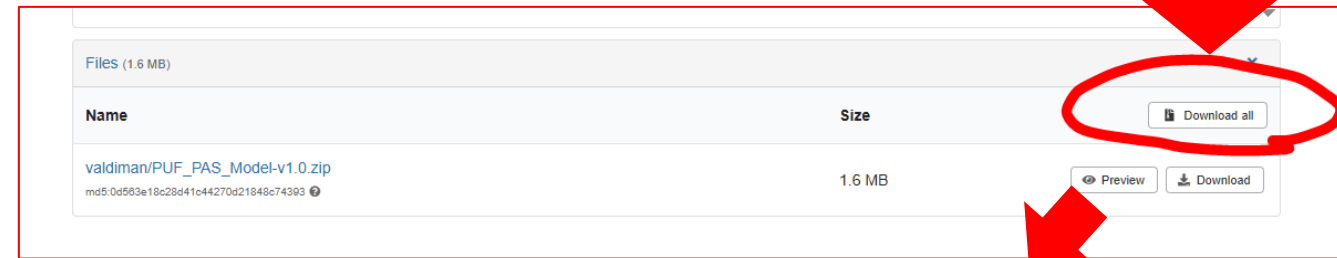
Item	Function
PUF_PAS_Model.Rproj	Opens the project with the correct working directory
Subfolders.R	Creates the required folder structure
R/Meteorology	Contains NOAA and MERRA meteorological scripts
R/PufPasEffectiveVolume	Contains NOAA and MERRA effective-volume models
Data	Contains PCB physicochemical-property tables
Output	Stores meteorological data and model results
README.md	Basic project instructions

Download and Extract the Project

- **Prerequisite:** Make sure R and RStudio Desktop are installed on your computer before starting
- Go to the **Zenodo repository:** <https://zenodo.org/records/14983867>
- Download the complete ZIP file to your computer
- Locate the downloaded ZIP file, right-click it, and select Extract All...
- Choose a location for the extracted folder and complete the extraction. You may rename the extracted folder for convenience.
- **Important:** always work from the extracted folder. Do not open or run the R scripts directly from inside the ZIP archive.
- Open RStudio Desktop to begin working with the model files

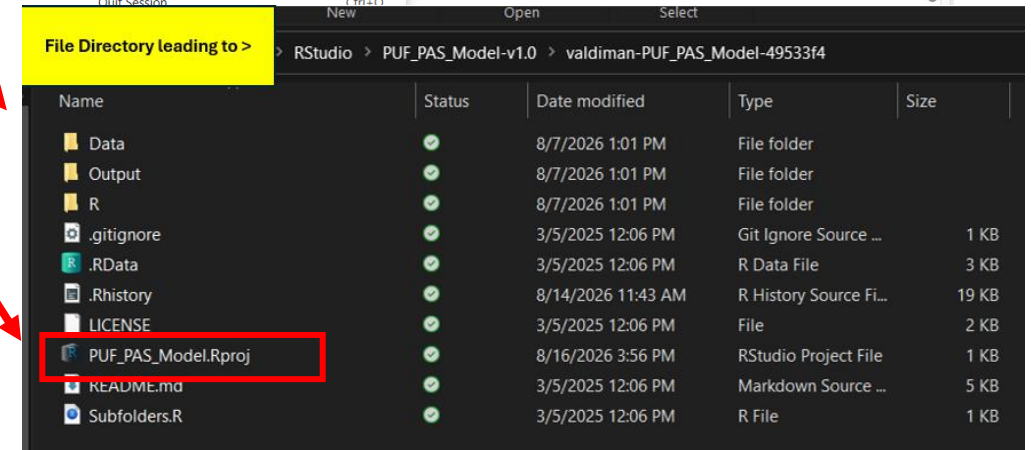
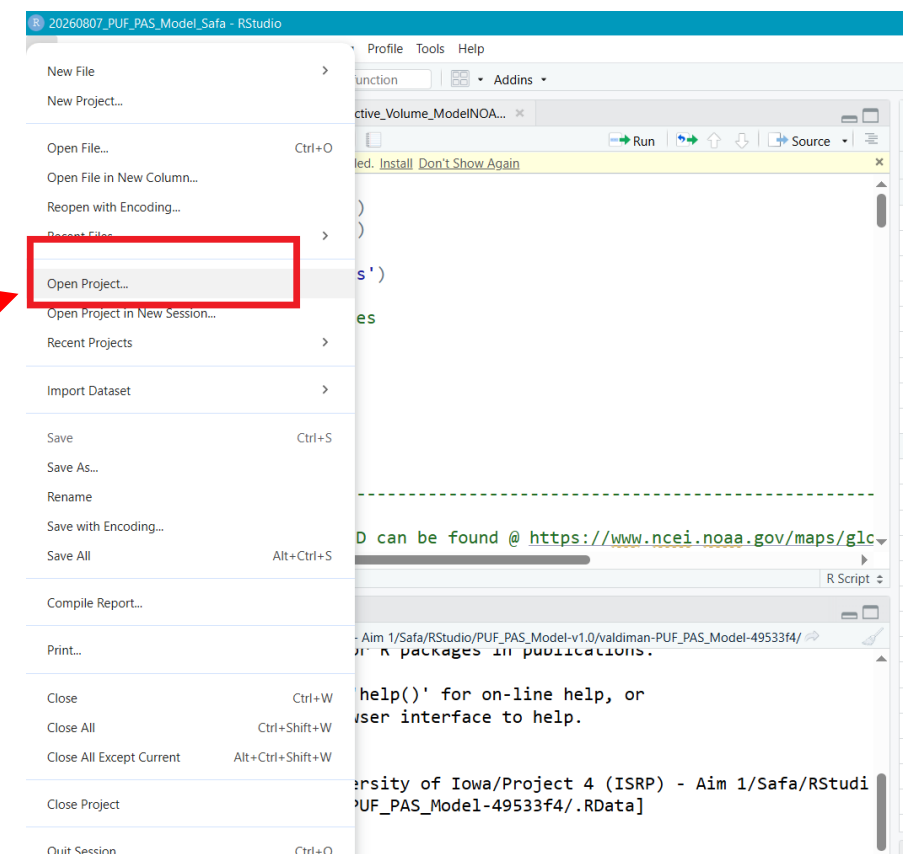


Scroll down



Open the Project in Rstudio

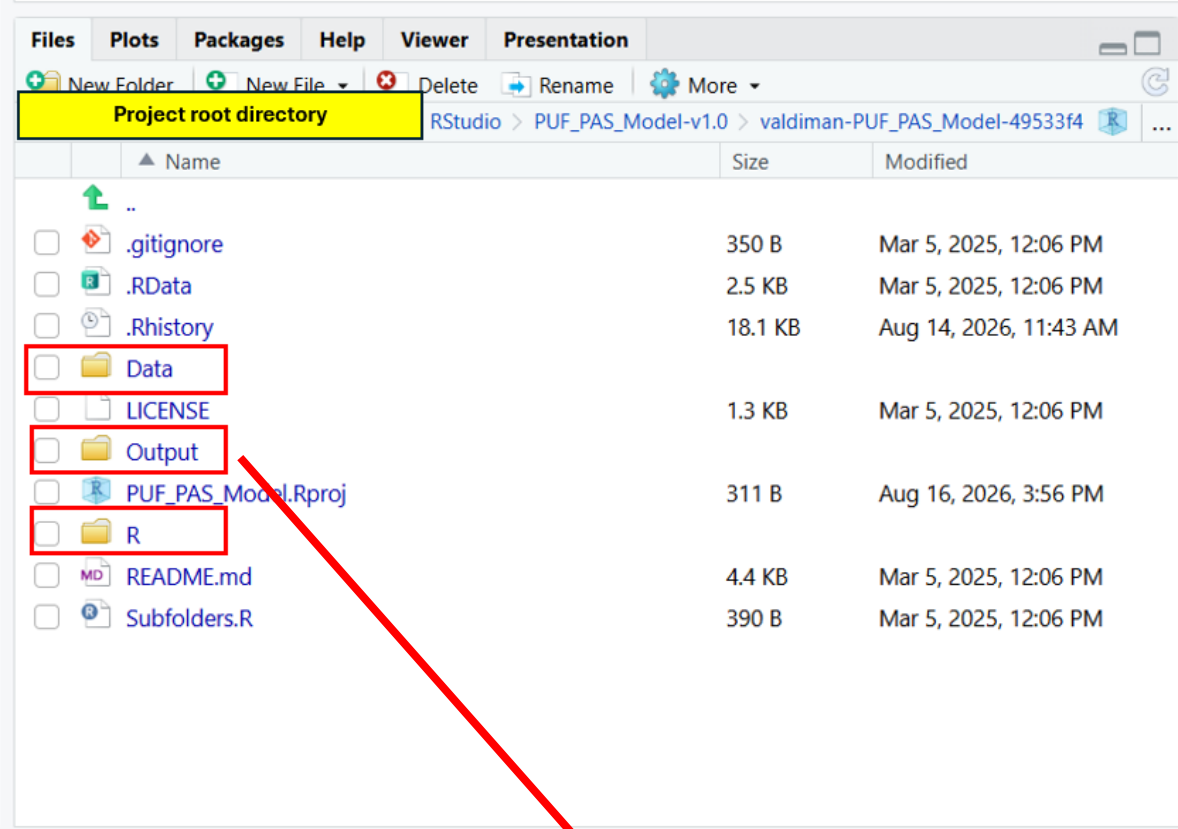
- Open RStudio Desktop
- **Select File → Open Project...**
- Navigate to the extracted project folder
- Select and open **PUF_PAS_Model.Rproj**
- The project will open in RStudio, with the project folder set as the working directory



Confirm that the Project Opened Correctly

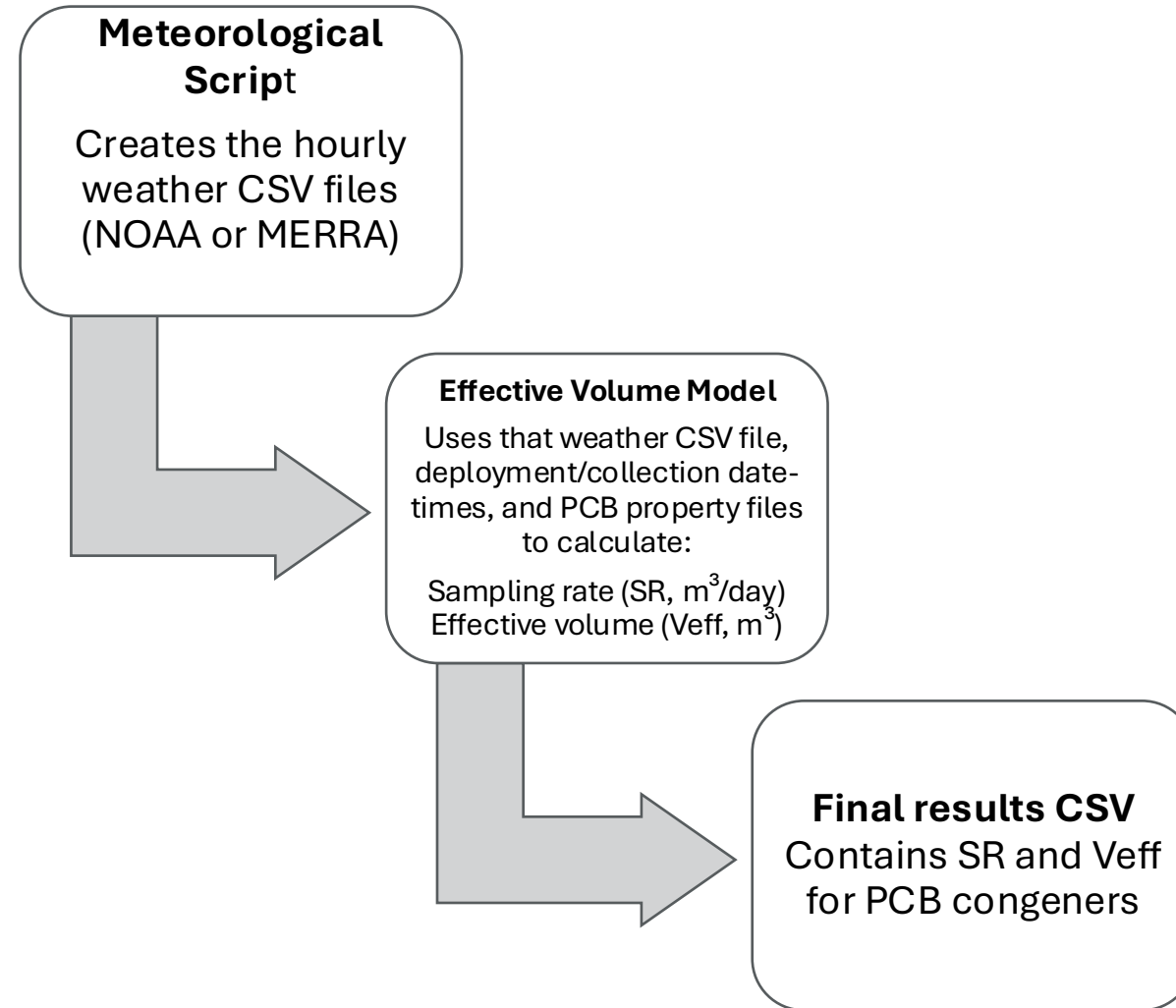
- The **Files** pane should display the project root containing “Data”, “Output”, and “R”

Item	Purpose
Data	PCB physicochemical-property input files
Output	Meteorological files and final model results
R	Meteorological-processing and effective-volume scripts
README.md	Project instructions
Subfolders.R	Creates the required folders if they are missing



Note: The CSV files already stored in **Output** are examples. Delete these example CSVs before running the scripts, but keep the Output folder and all its subfolders

Workflow



Choose NOAA or MERRA and gather the required inputs

Use NOAA when:

- A nearby representative **Global Hourly/ISD station** is available
- Its record covers the complete sampling period

Use MERRA when:

- No suitable nearby NOAA station is available.
- The required years are unavailable from the station.
- **MERRA inputs:** longitude/latitude and start/end dates.

NOAA required information	Purpose
Sampling location	Identify a nearby representative station
NOAA ISD station ID	Download hourly meteorological data
First sampling year	Set start_year
Number of calendar years	Set num_years
Deployment date and time	Define the beginning of model calculations
Collection date and time	Define the end of model calculations
Time zone	Convert deployment times to UTC

information and resources.

Search Options

Use the following search options to narrow down your results.

When ?

Start Date
mm/dd/yyyy

Select Date Range

Where ?

Ex: North Carolina

Iowa City

Station Search ?

Stations

Open Find-A-Station →

Global Hourly

Browse station files by map or file list, using any search options to narrow down your search.



Description

*This version has been superseded by a newer version. It is highly recommended for users to access the current version. Users should only access this superseded version for special cases, such as reproducing studies. If necessary, this version can be accessed by contacting NCEI.*The Integrated Surface Dataset (ISD) is composed of worldwide surface weather observations from over 35,000 stations, though the best spatial coverage is evident in North America, Europe, Australia, and parts of Asia. Parameters included are: air quality, atmospheric pressure, atmospheric temperature/dew point, atmospheric winds, clouds, precipitation, ocean waves, tides and more. ISD refers to the data contained within the digital database as well as the format in which the hourly, synoptic (3-hourly), and daily weather observations are stored. The format conforms to Federal Information Processing Standards (FIPS). ISD provides hourly data that can be used in a wide range of climatological applications. For some stations, data may go as far back as 1901, though most data show a substantial increase in volume in the 1940s and again in the early 1970s. Currently, there are over 14,000 "active" stations updated

Search Result Summary and List View

View a summary and list of your results based on the provided search options.

Result File Size

178.93 MB of 1.47 TB

Result File Count

39 of 709037

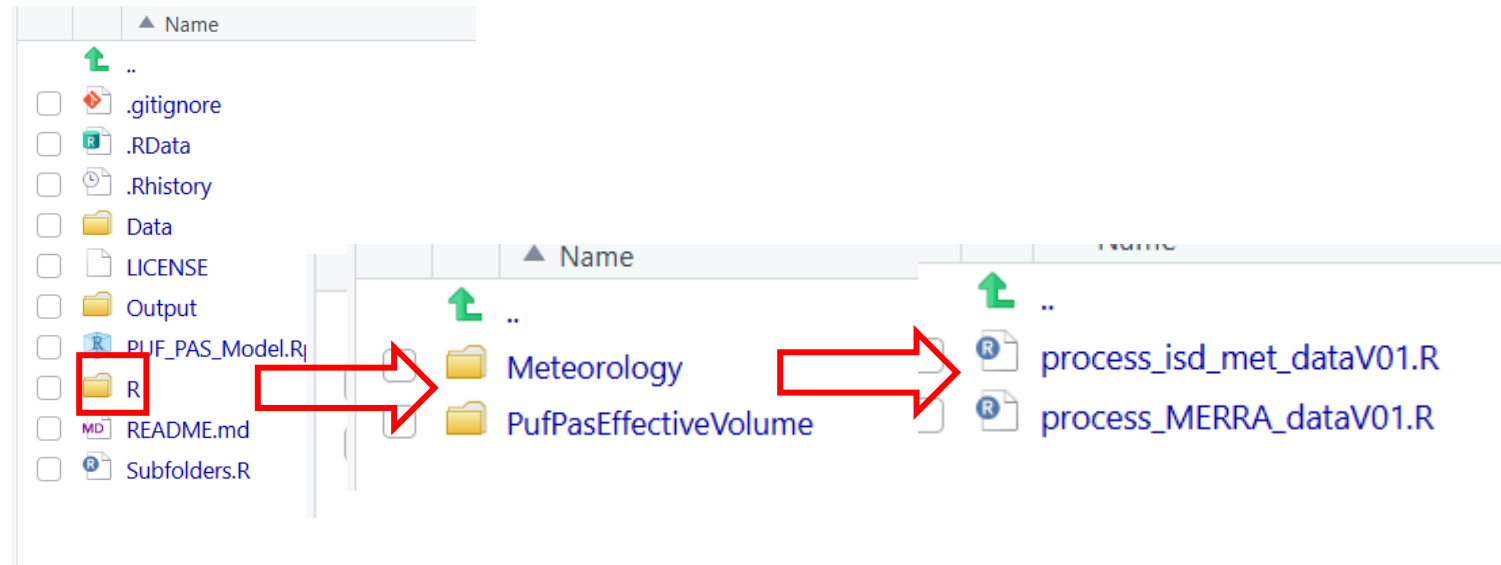
- <https://www.ncei.noaa.gov/access/search/datasets/global-hourly/>
- Search for the sampling city or location
- Select a nearby station
- Review the station's beginning and ending dates
- Confirm that both the deployment and collection dates fall within that record
- If they do not, check another nearby station or use MERRA

IOWA

6

Open the NOAA or MERRA Meteorological Script

- Open the meteorological data processing script
- In RStudio, go to the **Files pane**.
- Open the **R folder**, then the **Meteorology folder**
- Select the appropriate script based on the meteorological data source:
 - **NOAA data:**
process_isd_met_dataV01.R
 - **MERRA data:**
process_MERRA_dataV01.R
- Double-click the appropriate script to open it in the RStudio Source pane



NOAA Workflow: Enter the Station ID, Year & Number of Years

- Open “**process_isd_met_dataV01.R**”
- Install the required R packages if they are not already installed (lines 2— 5)
- Load the required libraries before running the data-processing code (lines 9 – 12)
- Press Ctrl + F, and search for: **metdataID** (line 32)
- Edit these three lines:
 - **metdataID** <- “XXXXXX-XXXXX”
 - **start_year** <- YYYY
 - **num_years** <- n

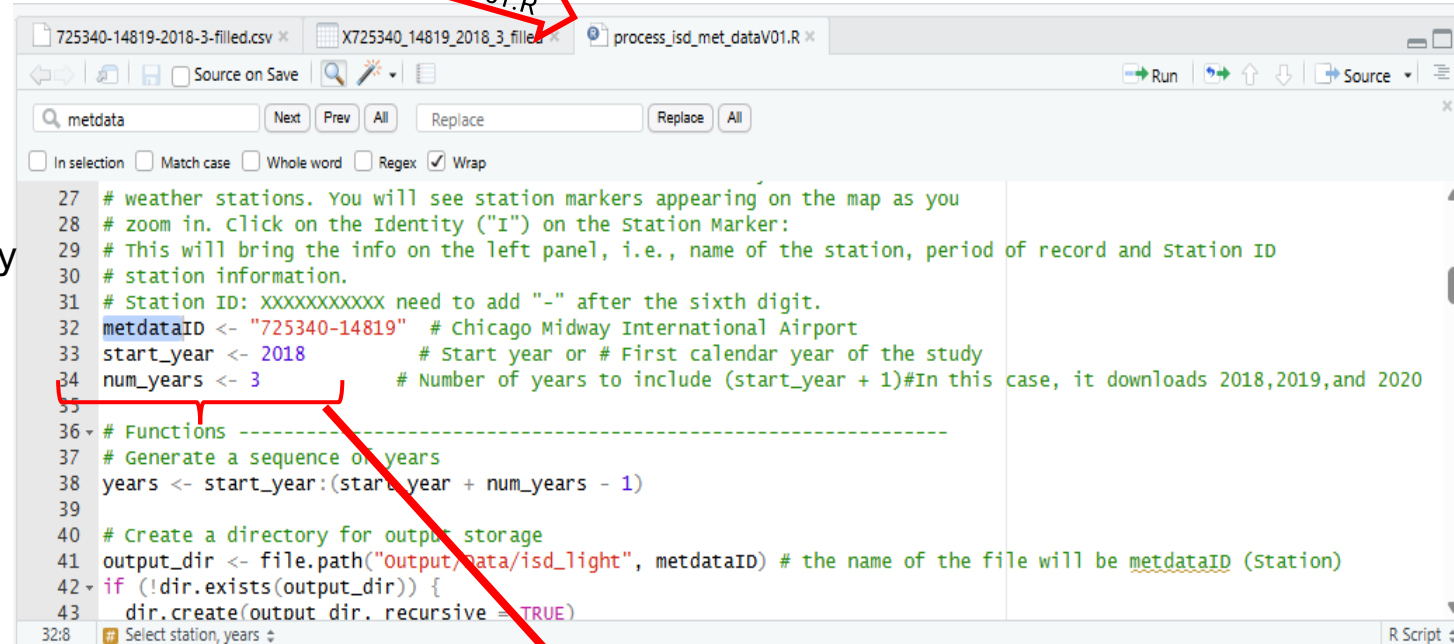
In this example:

metdataID = “725340-14819” (Chicago Midway Int. Airport)

start_year = 2018

num_years = 3 downloads 2018, 2019, and 2020

Run the script to process the meteorological data



```
27 # weather stations. You will see station markers appearing on the map as you
28 # zoom in. Click on the Identity ("I") on the Station Marker:
29 # This will bring the info on the left panel, i.e., name of the station, period of record and station ID
30 # station information.
31 # Station ID: XXXXXXXXXX need to add "-" after the sixth digit.
32 metdataID <- "725340-14819" # Chicago Midway International Airport
33 start_year <- 2018 # Start year or # First calendar year of the study
34 num_years <- 3 # Number of years to include (start_year + 1) In this case, it downloads 2018, 2019, and 2020
35
36 # Functions -----
37 # Generate a sequence of years
38 years <- start_year:(start_year + num_years - 1)
39
40 # Create a directory for output storage
41 output_dir <- file.path("output_data/isd_light", metdataID) # the name of the file will be metdataID (Station)
42 if (!dir.exists(output_dir)) {
43   dir.create(output_dir, recursive = TRUE)
44 }
```

Variable	Replace with
metdataID	Selected NOAA station ID
start_year	First calendar year required
num_years	Total number of calendar years to download
Example shown: Chicago Midway International Airport, 2018–2020	

Run the Meteorological Script (NOAA)

To run the code:

- If packages and libraries were already installed and loaded, no need to run it again
- You can run the script line by line by placing the cursor on a line and clicking **Run** (or pressing **Ctrl + Enter** on Windows / **Cmd + Enter** on Mac)
- Source will run the entire code
- You can also select multiple lines of code and click Run to execute all selected lines at once

1. Save the edited script using **Ctrl + S**

2. Click Source to run the entire script

On the first run, allow the required R packages to install

3. The script downloads hourly NOAA data for each selected year

4. Missing hourly values are identified and filled.

5. Confirm the final saved-file message

File directory

```
gunkzip
file_path <- file.path(output_dir, gsub(".*gz$", "", file_name))

The following objects are masked from 'package:methods':
  getClasses, getMethods

The following objects are masked from 'package:base':
  attach, detach, load, save

R.utils v2.13.0 (2025-02-24 21:20:02 UTC) successfully loaded. See ?R.utils for help.
Attaching package: 'R.utils'

The following object is masked from 'package:tidyr':
  extract

The following object is masked from 'package:utils':
  timestamp

The following objects are masked from 'package:base':
  cat, commandArgs, getopt, isopen, nullfile, parse, use, warnings

trying URL 'https://www1.ncdc.noaa.gov/pub/data/noaa/isd-11te/2018/725340-14819-2018.gz'
Content type 'application/gzip' length 90222 bytes (88 KB)
downloaded 88 KB

trying URL 'https://www1.ncdc.noaa.gov/pub/data/noaa/isd-11te/2019/725340-14819-2019.gz'
Content type 'application/gzip' length 90989 bytes (88 KB)
downloaded 88 KB

trying URL 'https://www1.ncdc.noaa.gov/pub/data/noaa/isd-11te/2020/725340-14819-2020.gz'
Content type 'application/gzip' length 90370 bytes (88 KB)
downloaded 88 KB

-----Data Quality Summary (Before Filling)-----
Total Missing Hours for TA = 1 (0.00%)
Total Missing Hours for TD = 0 (0.00%)
Total Missing Hours for WS = 2 (0.01%)
Total Missing Hours for Pr = 139 (0.53%)
Total Missing Hours for WD = 288 (1.09%)
Total Missing Hours for QV = 0 (0.00%)
Final filled data saved to output/Data/isd_light/725340-14819/725340-14819-2018-3-filled.csv
```

Environment	History	Connections	Tutorial
Global Environment			
metadataID			"725340-14819"
month			int [1:8783] 1 1 1 1 1 1 1 1 1 ...
num_years			3
output_dir			"output/Data/isd_light/725340-14819"
output_file_path			"output/Data/isd_light/725340-14819/725340-14819-2018-3-filled.csv"
Pr			num [1:8783] 101100 101130 101110 101110 101110 101110 101110 101110 101110 101110
start_date			2018-01-01 01:00:00 UTC
start_year			2018
TA			num [1:8783] -1.7 -1.7 -1.7 -1.7 -1.7 -2.8 -2.8 -2.8 -2.8 -2.8
TD			num [1:8783] -6.1 -5.6 -5.6 -5.6 -5.6 -5.6 -5.6 -5.6 -5.6 -5.6
WD			int [1:8783] 280 270 260 270 270 260 260 270 270 260
WS			num [1:8783] 6.7 5.7 7.7 4.1 7.7 3.6 4.1 3.6 4.1 3.6
Year			int [1:8783] 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020
years			int [1:3] 2018 2019 2020

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

File directory

- process_isd_met_dataV01.R
- process_MERRA_dataV01.R

Locate the Processed Meteorological File (NOAA)

- In RStudio's Files pane, navigate to (NOAA example):
 - **Output → Data → isd_light → 725340-14819**
- Locate the processed meteorological data file. In our example:
725340-14819-2018-3-filled.csv (3 means 2018, 2019 and 2020 combined)
- This CSV file contains the processed hourly meteorological data for the sampling location
- Note the location where the file was saved, as its file path will be entered later as “**met_data_path**” in the effective-volume model
- The output filename generally follows this format:
StationID-StartYear-NumberOfYears-filled.csv
- **For example**, 725340-14819-2018-3-filled.csv indicates Station ID 725340-14819, a start year of 2018, and 3 years of meteorological data

The screenshot illustrates the process of locating the processed meteorological file in RStudio. It shows the Files pane on the left with the path **Output → Data → isd_light → 725340-14819** highlighted by red arrows. The main window displays the contents of the file **725340-14819-2018-3-filled.csv**, which is a CSV table with 6 columns: **date**, **TA**, **Pr**, **WS**, **WD**, and **QV**. The table shows hourly data for the year 2018. A red box highlights the text "Viewing the output CSV files in RStudio".

	date	TA	Pr	WS	WD	QV
1	2018-01-01 01:00:00	259.85	103850	5.1	320	1.992565e-04
2	2018-01-01 02:00:00	258.75	103890	4.6	310	1.682531e-04
3	2018-01-01 03:00:00	257.55	103910	4.6	310	1.416051e-04
4	2018-01-01 04:00:00	257.55	103950	5.1	300	1.285761e-04
5	2018-01-01 05:00:00	256.45	103980	3.1	300	1.140827e-04
6	2018-01-01 06:00:00	255.95	103990	5.1	300	9.292015e-05
7	2018-01-01 07:00:00	254.85	104030	3.6	300	7.870054e-05
8	2018-01-01 08:00:00	254.25	104070	4.6	300	6.974542e-05
9	2018-01-01 09:00:00	253.75	104080	5.1	310	6.248060e-05
10	2018-01-01 10:00:00	253.15	104090	5.1	310	5.582380e-05
11	2018-01-01 11:00:00	252.55	104110	4.1	300	5.216917e-05
12	2018-01-01 12:00:00	252.05	104150	5.7	290	4.665611e-05
13	2018-01-01 13:00:00	251.45	104190	4.6	280	4.356413e-05
14	2018-01-01 14:00:00	251.45	104230	4.1	280	4.354741e-05
15	2018-01-01 15:00:00	252.05	104250	4.6	290	4.923508e-05
16	2018-01-01 16:00:00	253.15	104300	5.1	290	5.571140e-05
17	2018-01-01 17:00:00	254.25	104260	4.6	270	6.961832e-05
18	2018-01-01 18:00:00	255.95	104200	4.6	270	9.273289e-05
19	2018-01-01 19:00:00	256.45	104170	5.1	280	1.033374e-04
20	2018-01-01 20:00:00	257.05	104150	5.1	280	1.103229e-04
21	2018-01-01 21:00:00	257.05	104150	5.7	260	1.153150e-04
22	2018-01-01 22:00:00	256.45	104200	5.1	270	1.033076e-04
23	2018-01-01 23:00:00	255.35	104260	4.1	270	9.152984e-05
24	2018-01-02 00:00:00	254.85	104290	5.1	270	8.659391e-05
25	2018-01-02 01:00:00	253.75	104300	5.1	280	7.264392e-05

MERRA Workflow: Enter the Coordinates and Dates

- Open: “**process_MERRA_dataV01.R**”
- Install the required R packages if they are not already installed (lines 2 – 4)
- Load the required libraries before running the data-processing code (lines 8 – 10)
- Press **Ctrl + F** and search for “**lonlat**” (line 15)
- Replace the example coordinates with the coordinates of your sampling location. Enter the coordinates in longitude, latitude order. The coordinates can be obtained using **Google Earth**
- Replace the sampling years (yyyy-mm-dd)
- Search for **output_dir** (line 102) and replace the example site name in the output folder, and search for the **output_file_path** (line 108) and replace CSV filename (example: OHareAirport-2018-2020.csv)

Variable	Replace with
lonlat	Sampling-site longitude and latitude
start_date	First meteorological date required
end_date	Last meteorological date required
output_dir	Site-specific output folder name
output_file_path	Site- and period-specific CSV filename

```
5  
6 # Load necessary libraries  
7 {  
8   library(dplyr)  
9   library(nasapower)  
10  library(zoo)  
11 }  
12  
13 # Location and time selection  
14 # Define coordinates. E.g., O'Hare International Airport, Chicago (Longitude, Latitude)  
15 lonlat <- c(-87.904722, 41.978611)  
16  
17 # Define the start and end date  
18 start_date <- "2018-01-01"  
19 end_date <- "2020-12-31"  
20
```

```
100  
101 # Create output directory to store data  
102 output_dir <- file.path("Output/Data/MERRA", "OHareAirport") # Need to adjust name, e.g., O'Hare Airport  
103 if (!dir.exists(output_dir)) {  
104   dir.create(output_dir, recursive = TRUE)  
105 }  
106  
107 # Save the final filled data without 'T2MDEW'  
108 output_file_path <- file.path(output_dir, "OHareAirport-2018-2020.csv") # Need to adjust name  
109 tryCatch({  
110   write.csv(filled_data, file = output_file_path, row.names = FALSE)  
111   cat(sprintf("Final filled data saved to %s\n", output_file_path))  
112 }, error = function(e) {  
113   cat("Error saving the file:", e$message, "\n")  
114 })
```

Run the MERRA Meteorological Script

- To run the code:
- If packages and libraries were already installed and loaded, no need to run it again (see instruction on previous slide)
- Confirm the selected dates and coordinates in the **POWER** summary
- For this O'Hare 2018–2020 example, the processed dataset contains 26,304 hourly observations and six required variables
- Confirm completion using the **final saved-file** message

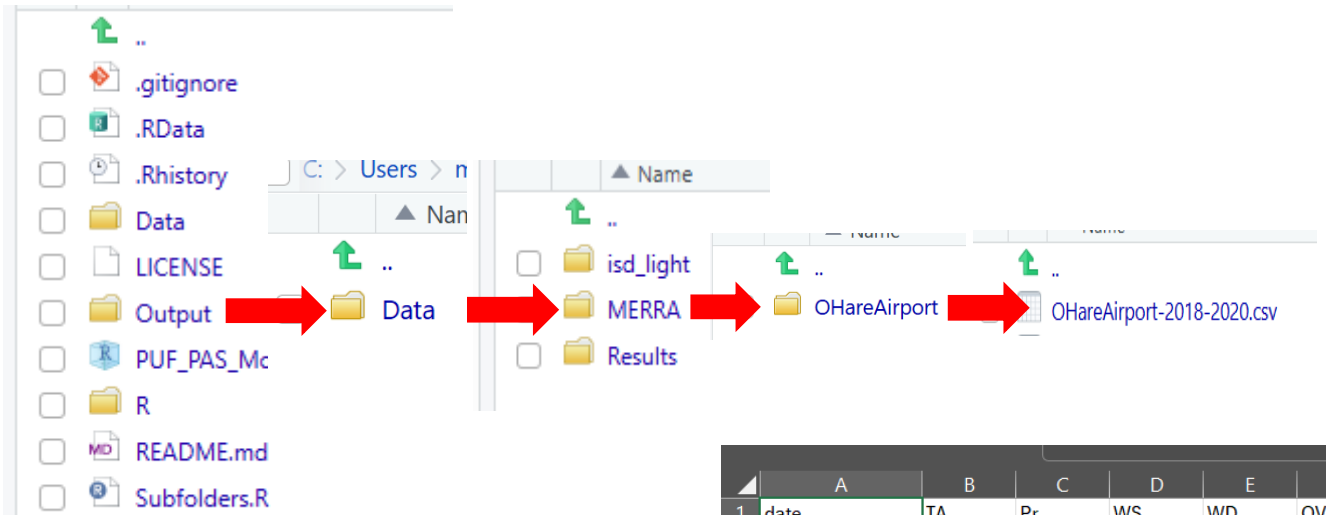
```
process_isd_met_dataV01.R × process_MERRA_dataV01.R × PUF_PAS_Effective_Volume_ModelNOA... ×
Source on Save
Packages nasapower and zoo required but are not installed. Install Don't Show Again
94:15 Download data from MERRA-2
R Script

Console Terminal Background Jobs
R 4.5.1 · G:/University of Iowa/Project 4 (ISRP) - Aim 1/Safa/RSudio/PUF_PAS_Model-v1.0/valdiman-PUF_PAS_Model-49533f4/

POWER.In [26,304 × 11] (S3: POWER.Info/tbl_df/tbl/data.frame)
$ LON : num [1:26304] -87.9 -87.9 -87.9 -87.9 -87.9 ...
$ LAT : num [1:26304] 42 42 42 42 42 ...
$ YEAR : num [1:26304] 2018 2018 2018 2018 2018 ...
$ MO : num [1:26304] 1 1 1 1 1 1 1 1 1 ...
$ DY : num [1:26304] 1 1 1 1 1 1 1 1 1 ...
$ HR : num [1:26304] 0 1 2 3 4 5 6 7 8 9 ...
$ T2M : num [1:26304] -16.7 -17.8 -18.7 -19.7 -20.5 ...
$ T2MDEW : num [1:26304] -19 -19.9 -20.7 -21.4 -22.1 ...
$ PS : num [1:26304] 101 101 101 101 101 ...
$ WD10M : num [1:26304] 316 314 312 311 310 ...
$ WS10M : num [1:26304] 5.97 6.06 6.18 6.3 6.26 6.3 6.38 6.25 6.11 6.15 ...
- attr(*, "spec")=
.. cols(
.. YEAR = col_double(),
.. MO = col_double(),
.. DY = col_double(),
.. HR = col_double(),
.. T2M = col_double(),
.. T2MDEW = col_double(),
.. PS = col_double(),
.. WD10M = col_double(),
.. WS10M = col_double()
.. )
- attr(*, "problems")=<externalptr>
- attr(*, "POWER.Info")= chr "NASA/POWER Source Native Resolution Hourly Data "
- attr(*, "POWER.Summary")= chr "Dates (month/day/year): 01/01/2018 through 12/31/2020 in UTC"
- attr(*, "POWER.Location")= chr "Location: Latitude 41.9786 Longitude -87.9047 "
- attr(*, "POWER.Elevation")= chr "Elevation from MERRA-2: Average for 0.5 x 0.625 degree lat/lon region = 231.39 meter s"
- attr(*, "POWER.Climate_zone")= chr "The value for missing source data that cannot be computed or is outside of the sou rces availability range: NA "
- attr(*, "POWER.Missing_value")= chr "Parameter(s): "
- attr(*, "POWER.Parameters")= chr "T2M MERRA-2 Temperature at 2 Meters (C) ;\n T2MDEW MERRA-2 Dew/Frost Poin t at 2 Meters (C) ;\n PS "| __truncated__
NULL
'data.frame': 26304 obs. of 6 variables:
$ date: POSIXct, format: "2018-01-01 00:00:00" "2018-01-01 01:00:00" ...
$ TA : num 256 255 254 253 253 ...
$ Pr : num 100740 100780 100810 100840 100880 ...
$ WS : num 5.97 6.06 6.18 6.3 6.26 6.3 6.38 6.25 6.11 6.15 ...
$ WD : num 316 314 312 311 310 ...
$ QV : num 1.55e-04 1.28e-04 1.08e-04 9.07e-05 7.76e-05 ...
Final filled data saved to Output/Data/MERRA/OHareAirport/OHareAirport-2018-2020.csv
```

Locate the Processed Meteorological file (MERRA)

- In RStudio's Files pane, navigate to (MERRA example): **Output → Data → MERRA → OHareAirport**
- Locate the processed meteorological data file. In our example:
OHareAirport-2018-2020.csv
- This CSV contains the processed hourly MERRA-2 meteorological data for the O'Hare 2018–2020 example
- Note the location where the file was saved, as its file path will be entered later as “**met_data_path**” in the MERRA effective-volume model
- MERRA filenames generally follow: SamplingLocation-StartYear-EndYear.csv; For example, “OHareAirport-2018-2020.csv”



Processed
MERRA-2
CSV
opened in
Excel

	A	B	C	D	E	F
1	date	TA	Pr	WS	WD	QV
2	1/1/2018	256.45	100740	5.97	315.8	0.000155
3	1/1/2018 1:00	255.39	100780	6.06	313.7	0.000128
4	1/1/2018 2:00	254.44	100810	6.18	312	0.000108
5	1/1/2018 3:00	253.49	100840	6.3	310.8	9.07E-05
6	1/1/2018 4:00	252.65	100880	6.26	309.8	7.76E-05
7	1/1/2018 5:00	251.98	100910	6.3	310.7	6.77E-05
8	1/1/2018 6:00	251.54	100930	6.38	311.3	6.18E-05
9	1/1/2018 7:00	251.13	100970	6.25	310.7	5.75E-05
10	1/1/2018 8:00	250.76	100990	6.11	308	5.43E-05
11	1/1/2018 9:00	250.28	100990	6.15	305	5.05E-05
12	1/1/2018 10:00	249.85	100990	6.25	302.1	4.74E-05
13	1/1/2018 11:00	249.55	101020	6.34	299.4	4.54E-05
14	1/1/2018 12:00	249.51	101060	6.5	299.2	4.51E-05
15	1/1/2018 13:00	249.92	101110	6.7	301.7	4.79E-05
16	1/1/2018 14:00	251.3	101150	7.25	305.5	5.97E-05
17	1/1/2018 15:00	252.58	101190	7.18	307.8	7.30E-05
18	1/1/2018 16:00	253.97	101190	6.82	307.8	9.14E-05
19	1/1/2018 17:00	255.67	101140	6.47	307.1	0.00012
20	1/1/2018 18:00	257	101070	6.15	303.9	0.000151
21	1/1/2018 19:00	257.88	101040	5.98	301.3	0.000176
22	1/1/2018 20:00	258.22	101050	5.83	299.6	0.000188
23	1/1/2018 21:00	257.62	101070	5.44	297.2	0.000176
24	1/1/2018 22:00	255.52	101090	4.74	292.3	0.000132
25	1/1/2018 23:00	254.1	101120	4.57	285.5	0.00011
26	1/2/2018	253.25	101130	4.58	280.6	9.90E-05

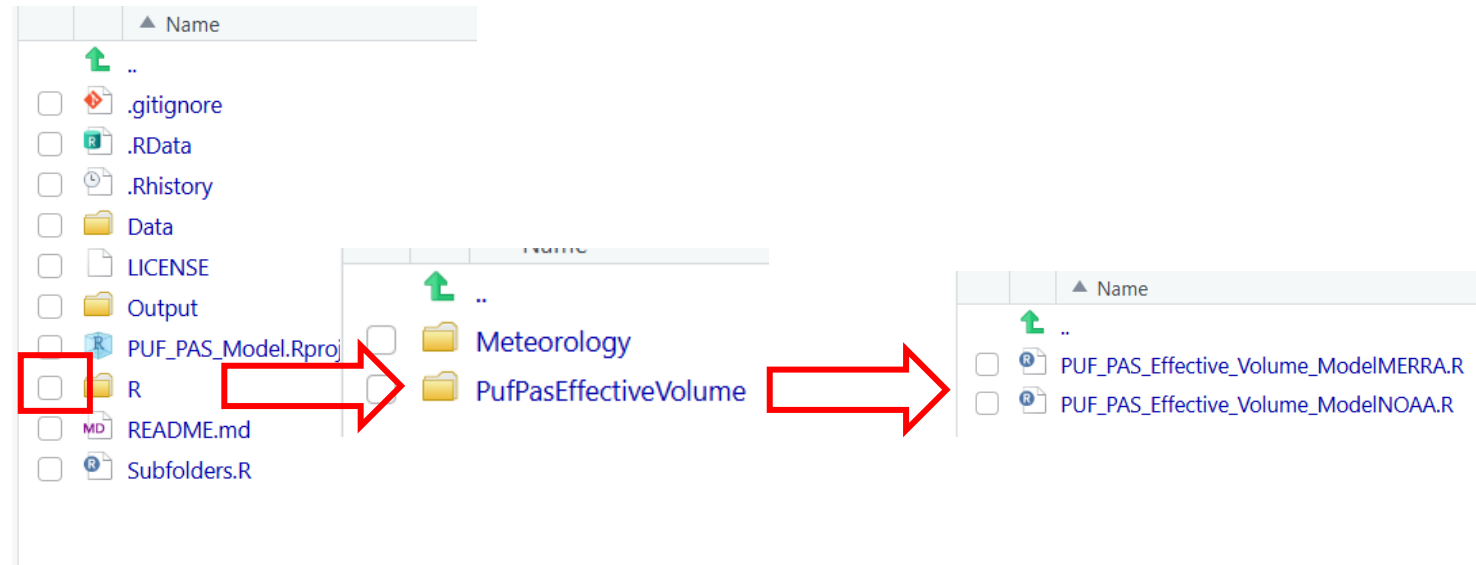
Open the Effective Volume Model Script (NOAA Or MERRA)

- In RStudio, go to the Files pane:

- Open the R folder, then open the **PufPasEffectiveVolume** folder
- Select the appropriate R script based on the meteorological data source:

- **NOAA data:**
PUF_PAS_Effective_Volume_ModelNOAA.R
- **MERRA data:**
PUF_PAS_Effective_Volume_ModelMERRA.R

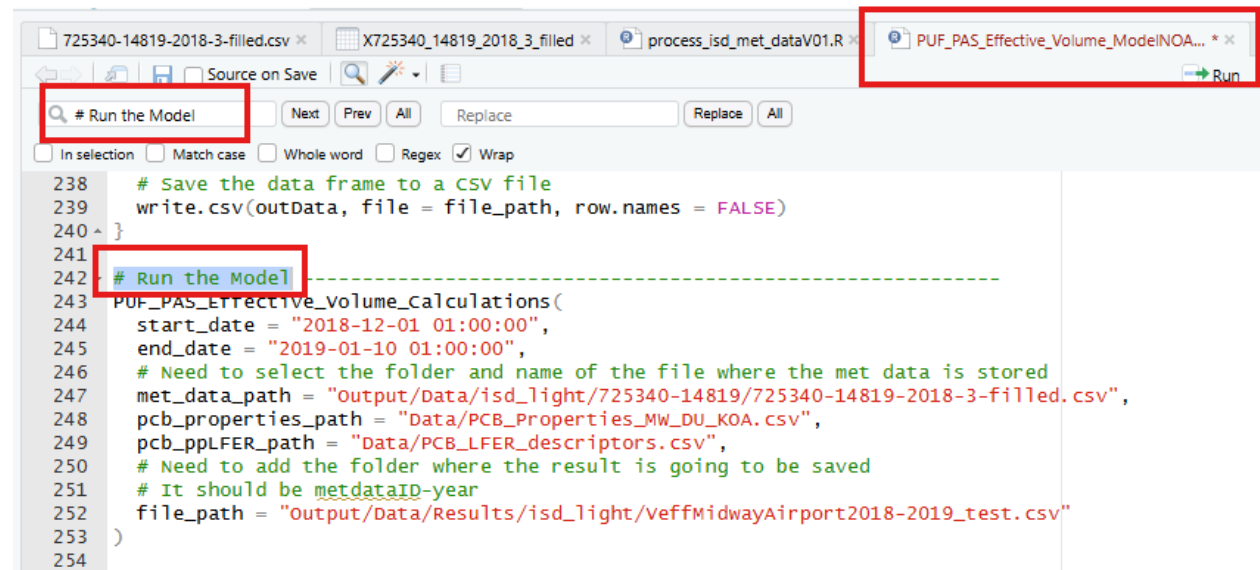
- Double-click the appropriate script to open it in the Rstudio Source pane



NOAA Effective Volume Model Script (PUF_PAS_Effective_Volume_ModelNOAA.R)

To run the code

- No additional R packages or libraries are required
- Run the function definitions (approximately lines 5–240)
 - These functions contain the calculations needed for the effective-volume model
 - You do not need to run each individual function separately; you can select and run the entire section at once
- Go to line 242, “**Run the Model.**” In this section:
 1. Enter the start and end dates of your deployment using the format **yyyy-mm-dd hh:mm:ss**. Enter 00 for minutes and seconds when they are not known or not needed. For example: 2018-12-01 01:00:00
 2. Check that “**met_data_path**” points to the correct location of your processed meteorological data (**line 247**)
 3. Check that **pcb_properties_path** points to the correct PCB properties file. We recommend using “**Data/PCB_Properties_MW_DU_KOA.csv**”.
 4. Check that **file_path** specifies the correct location and filename for the model output, under “**Output/Data/Results/isd_light/...**”

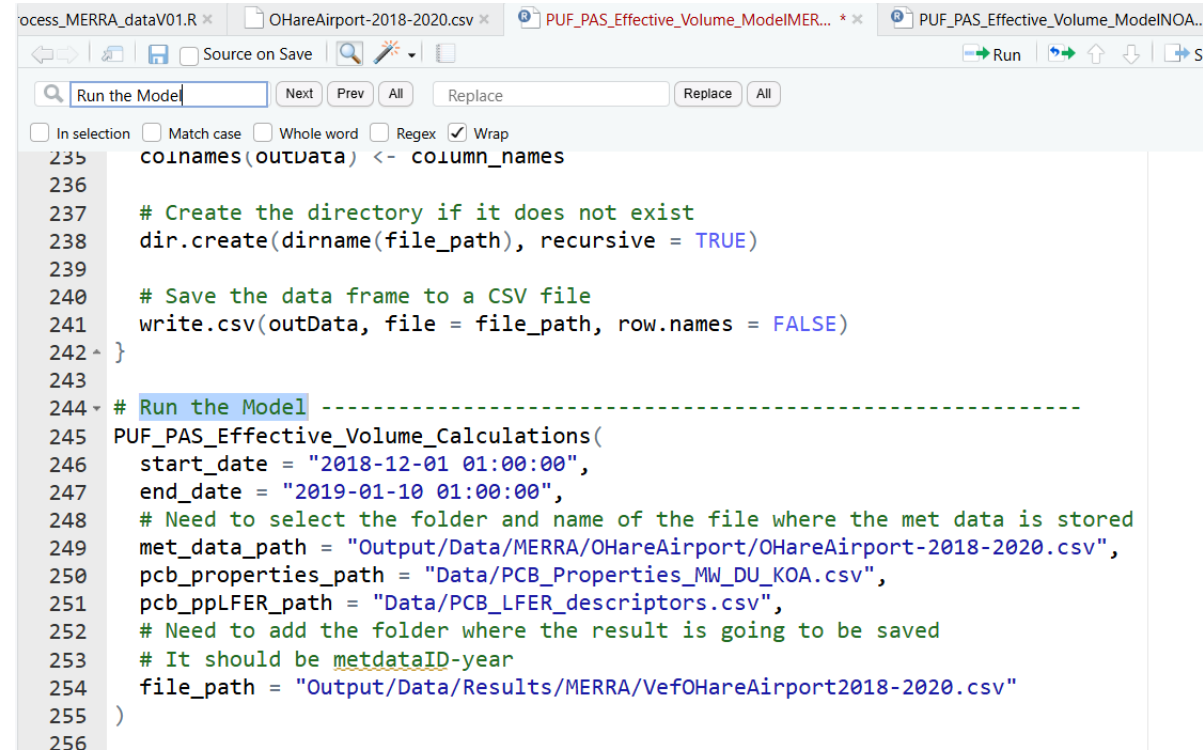


```
238 # Save the data frame to a csv file
239 write.csv(outData, file = file_path, row.names = FALSE)
240 }
241
242 # Run the Model
243 PUF_PAS_Effective_Volume_Calculations(
244   start_date = "2018-12-01 01:00:00",
245   end_date = "2019-01-10 01:00:00",
246   # Need to select the folder and name of the file where the met data is stored
247   met_data_path = "Output/Data/isd_light/725340-14819/725340-14819-2018-3-filled.csv",
248   pcb_properties_path = "Data/PCB_Properties_MW_DU_KOA.csv",
249   pcb_ppLFER_path = "Data/PCB_LFER_descriptors.csv",
250   # Need to add the folder where the result is going to be saved
251   # It should be metdataID-year
252   file_path = "Output/Data/Results/isd_light/VeffMidwayAirport2018-2019_test.csv"
253 )
254
```

MERRA Effective Volume Model Script (PUF_PAS_Effective_Volume_ModelMERRA.R)

To run the code

- No additional R packages or libraries are required
- Run the function definitions (approximately lines 6–242)
 - These functions contain the calculations needed for the effective-volume model
 - You do not need to run each individual function separately; you can select and run the entire section at once
- Go to line 244, “**Run the Model.**” In this section:
 1. Enter the start and end dates of your deployment using the format **yyyy-mm-dd hh:mm:ss**. Enter 00 for minutes and seconds when they are not known or not needed. For example: 2018-12-01 01:00:00
 2. Check that “**met_data_path**” points to the correct location of your processed meteorological data (line 249)
 3. Check that **pcb_properties_path** points to the correct PCB properties file. We recommend using “**Data/PCB_Properties_MW_DU_KOA.csv**”
 4. Check that **file_path** specifies the correct location and filename for the model output, under “**Output/Data/Results/MERRA/....**”



```
235 colnames(outData) <- column_names
236
237 # Create the directory if it does not exist
238 dir.create(dirname(file_path), recursive = TRUE)
239
240 # Save the data frame to a CSV file
241 write.csv(outData, file = file_path, row.names = FALSE)
242 }
243
244 # Run the Model -----
245 PUF_PAS_Effective_Volume_Calculations(
246   start_date = "2018-12-01 01:00:00",
247   end_date = "2019-01-10 01:00:00",
248   # Need to select the folder and name of the file where the met data is stored
249   met_data_path = "Output/Data/MERRA/OHareAirport/OHareAirport-2018-2020.csv",
250   pcb_properties_path = "Data/PCB_Properties_MW_DU_KOA.csv",
251   pcb_ppLFR_path = "Data/PCB_LFR_descriptors.csv",
252   # Need to add the folder where the result is going to be saved
253   # It should be metdataID-year
254   file_path = "Output/Data/Results/MERRA/VefOHareAirport2018-2020.csv"
255 )
256
```

Rerun the Model for Another Location and/or Deployment Period

To run the effective-volume model for another sampling location or deployment period:

- Make sure the meteorological data for the new location and dates have already been processed using the appropriate NOAA or MERRA script
- In the Run the Model section of the effective-volume script, update:
 - **start_date** - start date and time of the new deployment
 - **end_date** - end date and time of the new deployment
 - **met_data_path** - path to the processed meteorological data for the new location
 - **file_path** - path and filename where you want to save the new results
- Give each new output file a unique and descriptive filename. If you use the same filename and path as a previous run, the existing file will be overwritten. We recommend including information such as the sampling location and deployment dates in the filename to distinguish between model runs
- The PCB property files generally do not need to be changed
- Run the “**Run the Model**” section again
- Repeat these steps for each additional location and/or deployment period
- **Note:** If the function definitions are still loaded in your current R session, you do not need to run them again. If you restarted RStudio or started a new R session, run the function definitions before running the model
- Questions or additional information: Please contact Andres Martinez at andres-martinez@uiowa.edu